How (95)–(97) are obtained

a step-by-step companion to Sec. VI-E

every step below is checked by `analysis/vi_e_geometric_box.py`

0. What is being bounded, and why the rate

The estimator does not see the tilt angle. It fits the two-segment model to the *rate*

$$\omega_{\rm nom}(\tau) = C_1\big(\cosh C_2\tau - 1\big), \qquad C_1 = \frac{\dot M}{Wz_{CoM}}, \qquad C_2 = \sqrt{\frac{Wz_{CoM}}{J_P}}, \tag{D.1}$$

reconstructed from the gyro, and reports the onset as the argument that minimises the residual of that fit. Two questions follow, and (97) answers both:

(i) how far can the *true* rate depart from (D.1) before the fit stops describing the data? — the quantity $|e_\omega|_{\rm end}/\omega_{\rm nom,end}$;

(ii) how far does that departure move the *reported onset moment*? — the quantity $|\Delta M_{\rm crit}|$.

These are different questions with different answers, which is why the two halves of (97) carry different constants ($\frac{1}{2}$ and 1 in the first, $\frac{1}{7}$ and $\frac{1}{5}$ in the second). Section 5 answers (i) and Section 6 answers (ii). If only one line of this document is read, let it be that one.

The departure obeys the Duhamel integral (88),

$$e_\omega(\tau) = \int_0^\tau \dot h_\varphi(\tau - s)\,\rho(s)\,ds, \qquad \dot h_\varphi(u) = \frac{\cosh C_2 u}{J_P}, \tag{D.2}$$

with ρ the part of the true forcing that the estimator's span $\{1, \tau, \delta\varphi\}$ cannot absorb — the gravity remainder ρ_φ and the bilinear ground-effect coupling ρ_{GE} of (89). Everything below is about turning (D.2) into a number.

1. The crude bound, and why it is not enough

Put $|\rho(s)| \le \rho_{\max}$ inside (D.2) and pull it out:

$$|e_\omega(\tau_{\rm end})| \;\le\; \rho_{\max}\int_0^{\tau_{\rm end}} \dot h_\varphi(\tau_{\rm end} - s)\,ds \;=\; \rho_{\max}\int_0^{\tau_{\rm end}} \frac{\cosh C_2 u}{J_P}\,du \;=\; \frac{\rho_{\max}\sinh x}{J_P C_2}, \quad x \triangleq C_2\tau_{\rm end}. \tag{D.3}$$

The substitution $u = \tau_{\rm end} - s$ was used in the middle step; the integral of cosh is sinh. Using $J_P = Wz_{CoM}/C_2^2$, i.e. $J_P C_2 = Wz_{CoM}/C_2$, the inertia disappears: $|e_\omega(\tau_{\rm end})| \le \rho_{\max} C_2 \sinh x / Wz_{CoM}$. This is (90).

(D.3) is honest but blunt. It treats ρ as though it sat at its maximum for the whole window, whereas both channels start at zero at the onset and grow like τ^6 and τ^4. Evaluated over the geometric box of Sec. VI-D it returns a deviation of 446% at the slowest ramp — bigger than the signal it is supposed to bound. Sections 2–4 recover the lost factor.

2. Chebyshev's integral inequality

Statement. Let f, g be integrable on $[a, b]$. If they are *oppositely* monotone (one non-decreasing, the other non-increasing), then

$$\int_a^b fg \;\le\; \frac{1}{b-a}\int_a^b f \int_a^b g. \tag{D.4}$$

If they are monotone in the *same* direction, the inequality reverses. (Textbooks usually print the same-direction version, with $\ge$; the one needed here is the reversed one, so the direction is worth checking rather than remembering.)

Proof. Opposite monotonicity means $\big(f(u)-f(v)\big)\big(g(u)-g(v)\big) \le 0$ for every $u, v \in [a, b]$: when $u > v$ one bracket is ≥ 0 and the other ≤ 0. Integrate that product over the square $[a, b]^2$ and expand the four terms,

$$\underbrace{\iint f(u)g(u)}_{(b-a)\int fg} - \underbrace{\iint f(u)g(v)}_{\int f \int g} - \underbrace{\iint f(v)g(u)}_{\int f \int g} + \underbrace{\iint f(v)g(v)}_{(b-a)\int fg} \;=\; 2(b-a)\int fg - 2\int f \int g \;\le\; 0,$$

which is (D.4). □

Application to (D.2). Take $[a, b] = [0, \tau_{\text{end}}]$ with $f(s) = \rho(s)$ and $g(s) = \dot{h}_\varphi(\tau_{\text{end}} - s)$.

- f is non-decreasing: both channels vanish at the onset and grow monotonically over the window ($\rho_\varphi \propto \varphi^2$ with φ rising, $\rho_{GE} \propto \tau\varphi$ likewise). Equivalently, both are power series in u with non-negative coefficients, so $f' \ge 0$ term by term — the same sign pattern that Sec. 3 uses for a different purpose. This is the one place the argument needs the channels to increase; the reduction factors of Sec. 3 do not.
- g is non-increasing *in* s: as s grows the argument $\tau_{\text{end}} - s$ shrinks and cosh decreases. Note the kernel is increasing in its own argument; it is the reflection $s \mapsto \tau_{\text{end}} - s$ that makes it decreasing in s.

They are oppositely monotone, so (D.4) applies with $\le$, which is the useful direction:

$$e_\omega(\tau_{\text{end}}) \;\le\; \underbrace{\frac{1}{\tau_{\text{end}}}\int_0^{\tau_{\text{end}}} \rho}_{\bar{\rho}} \cdot \int_0^{\tau_{\text{end}}} \dot{h}_\varphi(\tau_{\text{end}} - s)\, ds \;=\; \frac{\bar{\rho}\sinh x}{J_P C_2}. \tag{D.5}$$

(D.5) is (D.3) with $\rho_{\max}$ replaced by the *time average* $\bar{\rho}$. Nothing else changed. All that remains is to compute $\bar{\rho}/\rho_{\max}$ for the two channels.

3. The reduction factors R_φ and R_{GE}: (95)

Both channels have a known shape, because φ does. From (D.1), $\varphi_{\text{nom}}(\tau) = (C_1/C_2)\Lambda(C_2\tau)$ with $\Lambda(u) = \sinh u - u$. Hence, writing $u = C_2\tau$,

$$\rho_\varphi(\tau) = \tfrac{1}{2}G''\varphi^2 = A\,\Lambda(u)^2, \qquad \rho_{GE}(\tau) = \beta_M \dot{M}\tau\,\varphi = B\,u\,\Lambda(u),$$

with A, B constants whose values never matter — they cancel in the ratio below.

Gravity channel. The maximum is at the window end, $\rho_{\varphi,\max} = A\Lambda(x)^2$. The average is, substituting $u = C_2\tau$ so that $d\tau = du/C_2$ and $\tau_{\text{end}} = x/C_2$,

$$\bar{\rho}_\varphi = \frac{1}{\tau_{\text{end}}}\int_0^{\tau_{\text{end}}} A\Lambda(C_2\tau)^2 d\tau = \frac{C_2}{x}\cdot\frac{A}{C_2}\int_0^x \Lambda(u)^2 du = \frac{A}{x}\int_0^x \Lambda(u)^2 du,$$

so the constant A cancels and

$$R_\varphi(x) \;\triangleq\; \frac{\bar\rho_\varphi}{\rho_{\varphi,\max}} \;=\; \frac{1}{x}\,\frac{\int_0^x \Lambda(u)^2 du}{\Lambda(x)^2}. \tag{D.6}$$

Ground-effect channel. Identically, $\rho_{GE,\max} = B\,x\Lambda(x)$ and

$$R_{GE}(x) \;=\; \frac{1}{x}\,\frac{\int_0^x u\,\Lambda(u)\,du}{x\,\Lambda(x)}. \tag{D.7}$$

How to read $1/7$ and $1/5$. The quickest way to see where these come from — not the proof, which is below — is the order of each channel. Keeping the leading term of $\Lambda(u) = u^3/6 + u^5/120 + \cdots$ makes $\Lambda^2 \simeq u^6/36$ and $u\Lambda \simeq u^4/6$, so the two channels are sixth and fourth order near the onset, and a τ^n signal averages over $[0,T]$ to $T^n/(n{+}1)$ against its endpoint value T^n: hence $\frac{1}{7}$ and $\frac{1}{5}$. At the other end $\Lambda \to \sinh u \simeq e^u/2$ gives $\int_0^x \Lambda^2 \simeq e^{2x}/8$ and $\int_0^x u\Lambda \simeq xe^x/2$, so $R_\varphi \simeq 1/(2x)$ and $R_{GE} \simeq 1/x$ — the origin of the $\frac{1}{2}$ and 1 that Sec. 5 needs, once $\Psi \simeq x$ multiplies them.

Neither reading is used as a proof. (D.6b) below obtains the small-x limits from l'Hôpital's rule with no expansion of f whatever, and Sec. 5 obtains the large-x ones the same way. Only the monotonicity argument that follows uses a series, and only for its sign pattern.

Why the small-x value is the supremum. Both R's decrease throughout, and this needs no numerics. Substituting $u = xt$ in the numerator rescales the window to $[0,1]$:

$$R(x) = \frac{1}{x}\frac{\int_0^x f(u)\,du}{f(x)} = \int_0^1 \frac{f(xt)}{f(x)}\,dt. \tag{D.6a}$$

Let

$$n_{\text{eff}}(u) \;\triangleq\; \frac{uf'(u)}{f(u)} \;=\; \frac{d\log f}{d\log u}$$

be the *local order* of f. The small-x limit of R follows from it by one application of l'Hôpital's rule, with no expansion of f: in the first form of (D.6a) numerator and denominator both vanish at $x = 0$, so

$$\lim_{x\to 0} R(x) = \lim_{x\to 0} \frac{\int_0^x f(u)\,du}{x\,f(x)} = \lim_{x\to 0} \frac{f(x)}{f(x) + xf'(x)} = \frac{1}{1 + n_{\text{eff}}(0^+)}, \tag{D.6b}$$

the last step dividing through by f. Only the limit of the local order is needed, never the coefficients of f. Differentiating the integrand of (D.6a) logarithmically,

$$\frac{\partial}{\partial x}\log\frac{f(xt)}{f(x)} = t\,\frac{f'(xt)}{f(xt)} - \frac{f'(x)}{f(x)} = \frac{n_{\text{eff}}(xt) - n_{\text{eff}}(x)}{x}.$$

For $t \in (0,1)$ we have $xt < x$, so if n_{eff} is increasing the bracket is ≤ 0: every integrand of (D.6a) decreases in x, hence so does R. $\square$

Both channels satisfy the hypothesis, and with $\Lambda' = \cosh u - 1$ the local orders are expressible through the same $\Psi_\varphi(u) = u(\cosh u - 1)/(\sinh u - u) = u\Lambda'/\Lambda$ that Sec. 4 discarded as the wrong normaliser:

$$f = \Lambda^2: \quad n_{\text{eff}} = 2\Psi_\varphi(u) \text{ from 6 to } \infty; \qquad f = u\Lambda: \quad n_{\text{eff}} = 1 + \Psi_\varphi(u) \text{ from 4 to } \infty.$$

Why n_{eff} increases. What is required here is easy to understate, so it is worth naming precisely. Since $n_{\text{eff}} = d\log f/d\log u$, the claim

$$\frac{dn_{\text{eff}}}{d\log u} = \frac{d^2(\log f)}{d(\log u)^2} \;\ge\; 0$$

says that $\log f$ is *convex* in $\log u$ — that the slope of f on log–log axes grows. Monotonicity of f gives only $n_{\mathrm{eff}} \geq 0$, which is the slope's *sign* and nothing more, and the two properties are independent: $f(u) = 1 - e^{-u}$, $u/(1{+}u)$ and $\log(1{+}u)$ are all strictly increasing and all have n_{eff} strictly *decreasing*, hence R increasing. A saturating signal has a higher average relative to its endpoint the longer the window, not a lower one. The two channels here behave the opposite way not because they rise but because they *stiffen*.

Convexity does hold, and one place a power series is used — only for its sign pattern, with no coefficient evaluated — establishes it.

Lemma (log–log convexity). Let $f(u) = \sum_{n\geq 0} a_n u^n$ converge for all $u \geq 0$ with every $a_n \geq 0$, at least two of them non-zero. Then $n_{\mathrm{eff}} = uf'/f$ is strictly increasing on $(0, \infty)$.

The second condition is what makes $f > 0$ for $u > 0$, so that dividing by f below is legitimate, and what makes the conclusion strict. No monotonicity of f is assumed — it would be redundant, since $f' = \sum_{n\geq 1} n a_n u^{n-1} \geq 0$ term by term, and it is not used in the proof.

Proof. Differentiating n_{eff},

$$u\,\frac{dn_{\mathrm{eff}}}{du} = \frac{uf'}{f} + \frac{u^2 f''}{f} - \Big(\frac{uf'}{f}\Big)^2 = \frac{S_0 S_2 - S_1^2}{S_0^2}, \qquad S_k \triangleq \sum_n n^k a_n u^n,$$

since $uf' = S_1$ and $u^2 f'' = S_2 - S_1$, so the first two terms combine to S_2/S_0. Every S_k runs from $n = 0$, including the two that come from differentiating: f' starts at $n = 1$ and f'' at $n = 2$, but the coefficients n and $n(n-1) = n^2 - n$ vanish on precisely the terms those shifts drop, so extending all three sums to the same index set neither adds nor loses anything. It is worth doing, because Cauchy–Schwarz is about to treat them as sequences over one index set, and those sequences are infinite: Λ, and with it Λ^2 and $u\Lambda$, has infinitely many non-zero coefficients. That is what the hypothesis of convergence for every $u \geq 0$ is for. A power series converging at some $u_0 > 0$ converges absolutely for $|u| < u_0$, so convergence at every $u \geq 0$ makes the radius infinite, and the termwise derivatives $\sum n a_n u^{n-1}$ and $\sum n(n{-}1) a_n u^{n-2}$ then converge on the same disc. Hence f'' exists as written and S_1, S_2 are finite — the two sequences lie in ℓ^2, which is what Cauchy–Schwarz for infinite sums requires.

Cauchy–Schwarz applied to $\{\sqrt{a_n u^n}\}$ and $\{n\sqrt{a_n u^n}\}$ — the square roots real because $a_n u^n \geq 0$, which is the only place the sign pattern is needed — gives $S_1^2 \leq S_0 S_2$. Hence $dn_{\mathrm{eff}}/du \geq 0$, with equality only when the two sequences are proportional, i.e. only when a single a_n is non-zero. □

(Dividing through by S_0 turns S_k/S_0 into the moments of the weights $a_n u^n/f$ and the numerator into their variance, which is a shorter way to say the same thing; nothing below depends on reading it that way.)

Both kernels qualify: $\Lambda = \sum_{k\geq 1} u^{2k+1}/(2k+1)!$ has non-negative coefficients, and so do the product Λ^2 and the shift $u\Lambda$; each has infinitely many non-zero, so certainly two. Hence n_{eff} is strictly increasing for both channels and R is strictly decreasing. Reading the $a_n u^n/f$ as weights also returns the origin values, since as $u \to 0$ they concentrate on the lowest exponent present, 6 and 4 — the same numbers the l'Hôpital route below obtains without any series.

The same fact as a closed-form derivative. An appendix that quotes dR/dx explicitly needs those expressions signed, and the argument above does it with no further machinery — in particular without the twelve-stage differentiation that signing them directly would cost. Write $A(x) = \int_0^x f$, so $R = A/(xf)$. Then, using $A' = f$ and $xf' = n_{\mathrm{eff}} f$,

$$R'(x) = \frac{A'\,xf - A\,(f + xf')}{x^2 f^2} = \frac{xf(x) - \big(1 + n_{\mathrm{eff}}(x)\big)A(x)}{x^2 f(x)}. \tag{D.7a}$$

The sign turns on whether A exceeds $xf/(1 + n_{\mathrm{eff}})$, and it does, for exactly the reason n_{eff}

increases. Since $d\log f/d\log u = n_{\rm eff}$ and $n_{\rm eff}(t) \le n_{\rm eff}(x)$ for $t \le x$,

$$\log\frac{f(x)}{f(u)} = \int_u^x n_{\rm eff}(t)\,\frac{dt}{t} \;\le\; n_{\rm eff}(x)\,\log\frac{x}{u} \quad\Longrightarrow\quad f(u) \;\ge\; f(x)\Big(\frac{u}{x}\Big)^{n_{\rm eff}(x)},$$

and integrating that minorant over $[0,x]$,

$$A(x) \;\ge\; f(x)\int_0^x \Big(\frac{u}{x}\Big)^{n_{\rm eff}(x)} du \;=\; \frac{x\,f(x)}{1+n_{\rm eff}(x)},$$

strictly, since $n_{\rm eff}$ is not constant. The numerator of (D.7a) is therefore negative and $R' < 0$. □

In words: f is at least as steep as the power law of its local order at x, so its average over the window is at least the $1/(n{+}1)$ that power law would give — and $1/(n{+}1)$ is precisely the value R would need for R' to vanish.

Substituting $n_{\rm eff} = 2x\Lambda'/\Lambda$ for $f = \Lambda^2$, and $n_{\rm eff} = (\Lambda + x\Lambda')/\Lambda$ for $f = u\Lambda$, and clearing denominators puts (D.7a) in the form an appendix would quote:

$$\frac{dR_\varphi}{dx} = \frac{x\Lambda^3 - (\Lambda + 2x\Lambda')\int_0^x \Lambda^2 du}{x^2\Lambda^3}, \quad \frac{dR_{GE}}{dx} = \frac{x^2\Lambda^2 - (2\Lambda + x\Lambda')\int_0^x u\Lambda\,du}{x^3\Lambda^2}, \tag{D.7b}$$

both negative for $x > 0$. The factors of x inside the brackets are load-bearing: with them the numerators cancel to order x^{12} and x^{10} at the origin, which is what leaves R' finite there; drop either and the expression diverges as x^{-2} instead. Both forms are checked against a five-point difference of R by `analysis/closed_form_check.py`.

The origin values without the series. The route above needs the coefficients' sign pattern; this one needs nothing, and is worth recording because it settles the same two numbers independently. The value of Ψ_φ at the origin is l'Hôpital's rule, applied three times. Writing $\Psi_\varphi = N/D$ with $N = u\cosh u - u$ and $D = \sinh u - u$, successive differentiation gives the four pairs

$$\frac{u\cosh u - u}{\sinh u - u}, \qquad \frac{\cosh u + u\sinh u - 1}{\cosh u - 1}, \qquad \frac{2\sinh u + u\cosh u}{\sinh u}, \qquad \frac{3\cosh u + u\sinh u}{\cosh u},$$

of which the first three are $0/0$ at $u = 0$ while the fourth is $3/1$. Hence $\Psi_\varphi(0^+) = 3$, so $n_{\rm eff}(0^+)$ is 6 for Λ^2 and 4 for $u\Lambda$, and (D.6b) returns $\frac{1}{7}$ and $\frac{1}{5}$. The mechanism is physical: Λ behaves as u^3 near the onset and grows exponentially later, so the larger the window the more sharply both channels are concentrated at its end, and the smaller their average is relative to the endpoint value. The small-x values are therefore suprema:

$$\bar\rho = R_\varphi(x)\,\rho_{\varphi,\max} + R_{GE}(x)\,\rho_{GE,\max}, \qquad R_\varphi \le \tfrac{1}{7}, \qquad R_{GE} \le \tfrac{1}{5}. \tag{95}$$

The suprema are approached but never attained: for $x > 0$ the inequalities are strict. Over the flown $x \in [1.79, 5.20]$ the factors run 0.089–0.133 and 0.145–0.191, so quoting $\frac{1}{7}$ and $\frac{1}{5}$ gives away 10–60% — while gaining a factor of 5 to 11 over (D.3).

4. Normalising on the rate: (96)

Divide (D.5) by the nominal rate at the same instant, $\omega_{\rm nom,end} = C_1(\cosh x - 1)$ with $C_1 = \dot M/Wz_{CoM}$, and use $J_P C_2 = Wz_{CoM}/C_2$:

$$\frac{|e_\omega|_{\rm end}}{\omega_{\rm nom,end}} \;\le\; \frac{\bar\rho\,C_2\sinh x/Wz_{CoM}}{(\dot M/Wz_{CoM})(\cosh x - 1)} \;=\; \frac{\bar\rho\,C_2}{\dot M}\cdot\frac{\sinh x}{\cosh x - 1}.$$

Wz_{CoM} has cancelled. Now eliminate $\dot M$ in favour of the moment increment the window actually spans. Since $\tau_{\rm end} = x/C_2$,

$$\Delta M_{\rm win} = \dot M\tau_{\rm end} = \frac{\dot M x}{C_2} \qquad\Longleftrightarrow\qquad \frac{C_2}{\dot M} = \frac{x}{\Delta M_{\rm win}},$$

and substituting,

$$\frac{|e_\omega|_{\rm end}}{\omega_{\rm nom,end}} \le \frac{\Psi(x)\,\bar\rho}{\Delta M_{\rm win}}, \qquad \Psi(x) = \frac{x\sinh x}{\cosh x - 1}. \tag{96}$$

The half-angle identities $\cosh x - 1 = 2\sinh^2(x/2)$ and $\sinh x = 2\sinh(x/2)\cosh(x/2)$ collapse the quotient:

$$\Psi(x) = x\,\frac{2\sinh(x/2)\cosh(x/2)}{2\sinh^2(x/2)} = x\coth\frac{x}{2} \;\ge\; 2.$$

Ψ rises from 2 at $x \to 0$ to x as $x \to \infty$. (Had the excursion been used as the normaliser instead of the rate, the factor would have been $x(\cosh x - 1)/(\sinh x - x)$, rising from 3 — a different function, and the wrong one, since the estimator never looks at φ.)

5. Why $\Psi\bar\rho$ stays bounded: first half of (97)

Ψ grows with x and the R's shrink with it, and the two products converge to $\frac{1}{2}$ and 1. Both bounds are exact, and neither needs the series: one corollary of the mean value theorem settles both, three applications for the first and one for the second.

The claims in integral form. Since $\Psi = x\coth(x/2)$ and $\rho_{\varphi,\max} = A\Lambda(x)^2$,

$$\Psi R_\varphi = x\coth\tfrac{x}{2}\cdot\frac{\int_0^x \Lambda^2 du}{x\Lambda(x)^2} = \coth\tfrac{x}{2}\,\frac{\int_0^x \Lambda^2 du}{\Lambda(x)^2},$$

so $\Psi R_\varphi \le \frac{1}{2}$ is exactly

$$\int_0^x \Lambda(u)^2 du \;\le\; \tfrac{1}{2}\tanh\tfrac{x}{2}\,\Lambda(x)^2, \tag{D.8}$$

and the same reduction turns $\Psi R_{GE} \le 1$ into

$$\int_0^x u\,\Lambda(u)\,du \;\le\; x\tanh\tfrac{x}{2}\,\Lambda(x). \tag{D.9}$$

The half-angle tangent is not an accident: $\tanh(x/2) = (\cosh x - 1)/\sinh x = \Lambda'(x)/\sinh x$, so both statements compare an integral of Λ against Λ and its own derivative.

Where $\frac{1}{2}$ and 1 come from. Both are the $x \to \infty$ limits, and each is one application of l'Hôpital's rule. Written as above, both products have the form $\coth(x/2)\int_0^x f/f(x)$ — with $f = \Lambda^2$ for the gravity channel and $f = u\Lambda$ for the ground-effect channel — and $\coth(x/2) \to 1$. In the remaining ratio numerator and denominator both diverge, so

$$\lim_{x\to\infty}\frac{\int_0^x f(u)\,du}{f(x)} = \lim_{x\to\infty}\frac{f(x)}{f'(x)} = \Big(\lim_{x\to\infty}\frac{f'}{f}\Big)^{-1},$$

and the two logarithmic derivatives are elementary:

$$\left.\frac{f'}{f}\right|_{f=\Lambda^2} = \frac{2\Lambda'}{\Lambda} = \frac{2(\cosh x - 1)}{\sinh x - x} \;\longrightarrow\; 2, \qquad \left.\frac{f'}{f}\right|_{f=u\Lambda} = \frac{1}{u} + \frac{\Lambda'}{\Lambda} \;\longrightarrow\; 1,$$

giving $\frac{1}{2}$ and 1. Each constant is thus the reciprocal of its kernel's exponential growth rate, which is what $\int_0^x e^{ku}du = e^{kx}/k$ says: the gravity channel gets the smaller constant because its kernel, $\Lambda^2 \sim e^{2x}/4$, is the square of the other's, $u\Lambda \sim u\,e^x/2$.

The only theorem used. Everything below rests on one corollary of the mean value theorem, applied four times in all — three for (D.8) and once for (D.9) — and nowhere supplemented.

Lemma (a corollary of the mean value theorem). If g is continuous on $[0, b]$, differentiable on $(0, b)$ with $g' \ge 0$ there, and $g(0) = 0$, then $g \ge 0$ on $[0, b]$. *Proof:* for $0 \le s < t \le b$ the mean value theorem gives $g(t) - g(s) = g'(\xi)(t-s) \geq 0$, so g is non-decreasing; taking $s = 0$ gives $g(t) \geq g(0) = 0$. $\square$ (The symbol is g, not the f of Sec. 3: this lemma is applied to ϕ', ϕ and D, never to a channel kernel.)

Note what is *not* used. No limit as $x \to \infty$ enters: the constant $\frac{1}{2}$ comes from the algebra of (D.8), and the inequality is established at each x separately. No monotonicity of ΨR_φ is assumed or proved. The limits quoted above serve only to show the constants cannot be improved.

Proof of (D.8). Let $D(x)$ be the right side minus the left, so that (D.8) is $D \geq 0$. Every quantity divided by below — $\coth(x/2)$, Λ^2, Λ, $\cosh^2(x/2)$ — is strictly positive for $x > 0$, so each step is an equivalence.

$D(0) = 0$, since $\tanh 0 = \Lambda(0) = \int_0^0 = 0$. Differentiating and factoring out $\Lambda > 0$,

$$D'(x) = \Lambda(x)\, B(x), \qquad B(x) = \tfrac{1}{4}\operatorname{sech}^2 \tfrac{x}{2}\, \Lambda + \tanh \tfrac{x}{2}\, \Lambda' - \Lambda,$$

so $D' \geq 0$ iff $B \geq 0$. Write $y = x/2$, $S = \sinh y$, $C = \cosh y$, giving $\Lambda = 2SC - 2y$ and $\Lambda' = 2S^2$. Multiplying B by $2C^2 > 0$,

$$2C^2 B = (SC - y) + 4S^3C - 4SC^3 + 4yC^2,$$

and since $S^3C - SC^3 = SC(S^2 - C^2) = -SC$ the two cubic terms collapse to $-4SC$, leaving $y(4C^2 - 1) - 3SC = y(3 + 4S^2) - 3SC$. Now $4S^2 = 2\cosh x - 2$ and $2SC = \sinh x$ and $y = x/2$, so

$$B(x) = \frac{\phi(x)}{4\cosh^2(x/2)}, \qquad \phi(x) \triangleq x + 2x\cosh x - 3\sinh x,$$

and $B \geq 0$ iff $\phi \geq 0$. Finally

$$\phi'(x) = 1 - \cosh x + 2x\sinh x, \qquad \phi''(x) = \sinh x + 2x\cosh x.$$

First application. $g = \phi'$: both terms of $\phi'' = \sinh x + 2x\cosh x$ are non-negative for $x \geq 0$, and $\phi'(0) = 1 - 1 + 0 = 0$. The Lemma gives $\phi' \geq 0$.

Second application. $g = \phi$: $\phi' \geq 0$ by the first application and $\phi(0) = 0$. The Lemma gives $\phi \geq 0$, hence $B \geq 0$, hence $D' = \Lambda B \geq 0$.

Third application. $g = D$: $D' \geq 0$ by the second application and $D(0) = 0$. The Lemma gives $D \geq 0$, which is (D.8). $\square$

Proof of (D.9). The same Lemma settles the ground-effect channel, and here it is needed only once. Let E be the right side of (D.9) minus the left,

$$E(x) \;=\; x\tanh \tfrac{x}{2}\, \Lambda(x) \;-\; \int_0^x u\, \Lambda(u)\, du, \qquad E(0) = 0.$$

Differentiating — the integral contributes $x\Lambda(x)$ — and using $\operatorname{sech}^2(x/2) = 2/(\cosh x + 1)$ and $\tanh(x/2) = \sinh x/(\cosh x + 1)$ to put everything over $\cosh x + 1$, the hyperbolic terms collapse to

$$E'(x) \;=\; \frac{\sinh^2 x - 2x\sinh x + x^2\cosh x}{\cosh x + 1} \;=\; \frac{\Lambda(x)^2 + x^2\Lambda'(x)}{\cosh x + 1}, \tag{D.10}$$

the second equality being the identity $(\sinh x - x)^2 + x^2(\cosh x - 1) = \sinh^2 x - 2x\sinh x + x^2\cosh x$. The numerator is a sum of two non-negative terms and the denominator is positive, so $E' \geq 0$; the Lemma gives $E \geq 0$, which is (D.9). $\square$

No series is needed for either, and (D.9) costs one application of the Lemma against three for (D.8) — the completed square in (D.10) does the work that ϕ did there.

No monotonicity of the products is claimed or needed — only that they do not exceed their limits, which is what (D.8) and (D.9) say. Expanding $\bar{\rho}$ by (95),

$$\Psi\bar{\rho} = \big(\Psi R_\varphi\big)\rho_{\varphi,\max} + \big(\Psi R_{GE}\big)\rho_{GE,\max} \;\leq\; \tfrac{1}{2}\rho_{\varphi,\max} + \rho_{GE,\max},$$

which is the left half of (97). *x has disappeared* — and with it C_2, J_P and Wz_{CoM}. This is the whole point of the exercise: the bound no longer depends on the window length, so it does not have to be solved for, only bounded well enough to keep $\rho_{GE,\max}$ finite (which is what (92)–(93) do).

6. From the rate deviation to the threshold: second half of (97)

The reported quantity is not e_ω but the onset moment, so the deviation must be pushed one step further.

How the onset responds to a perturbation. The fit is two-segment and *both* segments are in the cost — a constant baseline before the onset, the closed-form rise after it, sharing that baseline:

$$J(t_c) = \int_{t_a}^{t_c} \big(y - C_0\big)^2 dt + \int_{t_c}^{t_{\text{end}}} \Big(y - C_0 - C_1\big(\cosh C_2(t-t_c) - 1\big)\Big)^2 dt,$$

with C_0 the pre-onset baseline, refitted at each candidate t_c from the samples to its left. In the shifted variable $\tau = t - t_c$ the post-onset window is $[0, T]$ with $T = t_{\text{end}} - t_c = \tau_{\text{end}}$, the window Sec. 2 averages over, so the $\bar\rho$ below is the same $\bar\rho$.

Write the data as the true pre-onset level, the nominal response from the true onset, and a deviation,

$$y(t) = b + \hat\omega(t; t_c^*) + e_\omega(t), \qquad t_c = t_c^* + \delta t_c,$$

the perturbation being ρ itself and not its average. Carrying b rather than setting it to zero is what later gives C_0 something to estimate; without it C_0 would enter the cost and leave the stationarity condition unaccounted for.

One point about that definition is worth making explicitly, because getting it wrong invalidates everything after it. *e_ω is anchored at the true onset t_c^*, not at the candidate.* It is the difference between the recorded rate and the nominal solution of the true problem, so once the run is recorded it is a fixed function of t and

$$\frac{\partial e_\omega}{\partial t_c} = 0 \qquad \text{identically.}$$

When the residual is written $e_\omega - \delta t_c\,\chi - C$ below, every appearance of the optimisation variable is in δt_c and in C; none is in e_ω. Reading e_ω instead as "the residual at the current candidate" would make it depend on t_c and would count the same dependence twice.

Name the sensitivity

$$\chi(t) \;\triangleq\; \frac{\partial\hat\omega(t;t_c)}{\partial t_c}\bigg|_{t_c=t_c^*} \;=\; -\,C_1C_2\sinh\!\big(C_2(t-t_c^*)\big).$$

It needs a symbol of its own, and not φ: that is the tilt angle everywhere else, and $\langle e_\omega, \varphi\rangle$ would read as a rate correlated against an angle. To first order $\hat\omega(t; t_c^* + \delta t_c) \simeq \hat\omega(t;t_c^*) + \delta t_c\,\chi$, so the post-onset residual is $e_\omega - \delta t_c\,\chi - (C_0 - b)$: the baseline enters only through its *error*, a point taken up below. Setting that error aside for the moment,

$$\frac{\partial J}{\partial t_c} = \underbrace{\big(y(t_c)-C_0\big)^2 - \big(y(t_c)-C_0\big)^2}_{=\,0,\ \text{the two limits}} - 2\int_{t_c}^{t_{\text{end}}} (e_\omega - \delta t_c\,\chi)\chi\,dt \;-\; 2\frac{dC_0}{dt_c}\int r \;=\; 0.$$

t_c is the upper limit of one integral and the lower limit of the other, so it produces two boundary terms, and they cancel *identically*: at $\tau = 0$ the rise contributes $C_1(\cosh 0 - 1) = 0$, so the two-segment model is continuous at the onset by construction and both terms are $\big(y(t_c) - C_0\big)^2$ with opposite signs. Nothing is discarded and no order argument is needed. This is worth stating, because the first thing to check about a fit whose window endpoint is also its optimisation variable is whether the Leibniz terms were quietly dropped.

The baseline still couples the two segments, since C_0 depends on t_c through the samples it is fitted to. Its contribution to the pre-onset integral is $(dC_0/dt_c)\int_{t_a}^{t_c}(y - C_0)$, which vanishes — for the least-squares constant by its normal equation, and for the median the code actually uses because a median is piecewise constant in the sample set, so $dC_0/dt_c = 0$ almost everywhere. What could survive is the same drift entering the post-onset residual, but $dC_0/dt_c = \big(y(t_c) - C_0\big)/(t_c - t_a)$ for the mean, and at the onset the rate is still the baseline, so it vanishes there too. Measured across the 140 runs, C_0 moves by under 7.4×10^{-4} rad/s when the candidate onset shifts by one sample. Dropping that term and solving,

$$\delta t_c = \frac{\langle e_\omega, \chi\rangle}{\|\chi\|^2}, \qquad \langle a, b\rangle \triangleq \int_{t_c^*}^{t_{\text{end}}} a(t)\, b(t)\, dt, \tag{D.11}$$

the division legitimate because $\|\chi\|^2 > 0$ — $\sinh(C_2\tau)$ is not identically zero on a window of positive length.

Where the second derivative of $\hat{\omega}$ went. Only χ appears in the stationarity condition because $\partial J/\partial t_c$ differentiates the residual once; $\partial_{t_c}\chi$ belongs to the *second* derivative of J, and (D.11) is the Gauss–Newton form that drops it. Writing the one-parameter Newton step exactly,

$$J'(t_c^*) = -2\langle e_\omega, \chi\rangle, \qquad J''(t_c^*) = 2\|\chi\|^2 - 2\langle e_\omega, \partial_{t_c}\chi\rangle,$$

$$\delta t_c = -\frac{J'}{J''} = \frac{\langle e_\omega, \chi\rangle}{\|\chi\|^2 - \langle e_\omega, \partial_{t_c}\chi\rangle}, \qquad \partial_{t_c}\chi = C_1 C_2^2 \cosh(C_2\tau).$$

The neglected term is the residual times a curvature, the one nonlinear least squares always discards. It is legitimate here for a reason worth stating plainly: *the expansion parameter is e_ω, not δt_c*, and δt_c is itself first order in e_ω by (D.11). So $\langle e_\omega, \chi\rangle$ and $\delta t_c\|\chi\|^2$ are both $\mathcal{O}(e_\omega)$ while $\delta t_c\langle e_\omega, \partial_{t_c}\chi\rangle$ is $\mathcal{O}(e_\omega^2)$ — every term carrying a derivative of χ picks up an extra factor of the deviation, including the $(\delta t_c)^2\partial_{t_c}\chi$ left over from expanding the residual itself.

Solving for δt_c directly. Once the residual is linearised, J is an *exact* quadratic in δt_c, so it is cleaner to differentiate in that variable and solve for the value that zeroes the derivative. Since $t_c = t_c^* + \delta t_c$ the operator is the same, but the two approximations then sit in separate places: linearising the residual is the only one, and the normal equation that follows is exact for the quadratic it is applied to.

What the baseline has to satisfy, and what it does not. Keeping C_0 inside that square matters. Differentiating $(e_\omega - \delta t_c\chi - C_0)^2$ and solving gives

$$\delta t_c = \frac{\langle e_\omega - C_0,\ \chi\rangle}{\|\chi\|^2} = \frac{\langle e_\omega, \chi\rangle}{\|\chi\|^2} \;-\; C_0\,\frac{\langle 1, \chi\rangle}{\|\chi\|^2}, \tag{D.11a}$$

and the second term does not vanish: χ is of one sign, so $\langle 1, \chi\rangle = -C_1(\cosh x - 1) \neq 0$.

It does not follow that C_0 must be driven to zero, which is the natural but wrong reading of the onset condition. Only the *error* of C_0 against the true pre-onset level b survives: a run whose gyro sits at $b = 0.2$ rad/s is displaced by nothing at all provided $C_0 = b$, while a mismatch of 0.001 rad/s on a run with $b = 0$ displaces the onset measurably. What is required is an unbiased estimate of the pre-onset level — which is why the estimator fits C_0 rather than fixing it, and fits a median: the pre-onset gyro noise is heavy-tailed, and a candidate that overshoots leaks the rise into the segment being averaged.

Carrying (D.11a) through $\Delta M_{\text{crit}} = \dot{M}\,\delta t_c$ and $\|\chi\|^2 = C_1^2 C_2\left[\frac{1}{4}\sinh 2x - \frac{1}{2}x\right]$, the amplitude cancels as everywhere else and

$$\Delta M_{\text{crit},C} = Wz_{CoM}\,(C_0 - b)\,\frac{\cosh x - 1}{C_2\left[\frac{1}{4}\sinh 2x - \frac{1}{2}x\right]}. \tag{D.11b}$$

Minimising the cost directly reproduces (D.11b) to three figures. Over the box the gain runs from 0.042 N m per rad/s at the slowest ramp to 0.351 at the fastest, the fast end being worse because the window is shorter. The baselines the sweep actually forms — medians over the excitation window up to the candidate, not over the whole quiet record before it — are 3.5 mrad/s in the median and 47.8 at worst across the 140 runs, so bounding $|C_0 - b|$ by $|C_0|$ gives 0.054 mm of offset in the median and 0.558 mm at worst.

That bound is loose, and the direct measurement says so. Rerunning the whole pipeline with C_0 pinned to zero instead of fitted, which is the largest perturbation this term admits, moves the direction-paired M_{ff} by 0.80 mN m in the median and 4.91 at worst — 0.027 mm and 0.163 mm. The pairing cancels most of what the bound forwards. Neither statistic is better: the ramp-invariance CV is 0.0287 against 0.0292, a paired t of 0.87 on 19 degrees of freedom, and 66 of the 140 runs return an identical onset index either way. The median is kept because a pinned zero has no defence against genuine pre-onset motion, not because it measures better.

Numerically, $\langle e_\omega, \partial_{t_c}\chi\rangle/\|\chi\|^2$ is 9.05×10^{-5}, 9.05×10^{-4}, 9.05×10^{-3} and 9.05×10^{-2} as the deviation grows through four decades: exactly proportional to it, as an $\mathcal{O}(e_\omega)$ relative error must be. That ratio is the whole of the discrepancy reported below. With $\Delta M_{\text{crit}} = \dot{M}\,\delta t_c$ and $C_1 = \dot{M}/Wz_{CoM}$ the amplitude C_1 cancels between numerator and denominator, leaving

$$\Delta M_{\text{crit}} = -\,\frac{Wz_{CoM}\,\langle e_\omega,\ \sinh(C_2\tau)\rangle}{C_2\,\|\sinh(C_2\tau)\|^2}. \tag{D.12}$$

This is exactly the reduction `error_budget.py` implements, and it is the manuscript's (98)–(106) written with χ named: (D.11) is its (104), the baseline term below is its (105)–(106). Minimising J numerically against a deviation of the true shape, over four decades of deviation size, gives ratios to the formula of 0.99995, 0.99953, 0.99530 and 0.95559: it is the first-order term, as claimed. Minimising the full two-segment cost and the post-onset term alone gives the same answer to five figures at every size, which is the numerical form of the two paragraphs above.

From (D.12) to (97b), step by step. The reduction is short but has three distinct moves in it, so they are separated here. Throughout, τ runs from 0 at the onset to τ_{end} at the window end, and $\langle a, b\rangle = \int_0^{\tau_{\text{end}}} ab\,d\tau$.

Step 1: write the inner product out. (D.12) reads

$$\Delta M_{\text{crit},e} = -\,\frac{Wz_{CoM}}{C_2}\,\frac{\int_0^{\tau_{\text{end}}} e_\omega(\tau)\sinh(C_2\tau)\,d\tau}{\|\sinh(C_2\tau)\|^2}.$$

Step 2: substitute the deviation. e_ω is not free: (D.2) gives it as the Duhamel integral of the forcing, with $\dot{h}_\varphi(u) = \cosh(C_2 u)/J_P$ from (87). Putting that in makes the numerator a double integral over the triangle $0 \le s \le \tau \le \tau_{\text{end}}$,

$$\int_0^{\tau_{\text{end}}} \sinh(C_2\tau)\left[\int_0^{\tau} \frac{\cosh\big(C_2(\tau - s)\big)}{J_P}\,\rho(s)\,ds\right] dt.$$

Step 3: exchange the order. Fubini applies — both integrands are continuous on a bounded domain — and the same triangle read the other way has s outermost with τ running from s up to τ_{end}:

$$= \int_0^{\tau_{\text{end}}} \rho(s)\,\underbrace{\left[\frac{1}{J_P}\int_s^{\tau_{\text{end}}} \sinh(C_2\tau)\cosh\big(C_2(\tau - s)\big)\,d\tau\right]}_{\text{a function of } s \text{ alone}} ds.$$

The forcing is now outside, multiplied by something that no longer depends on it. That is the whole point of the step: what began as a correlation of e_ω against a kernel has become a weighted average of ρ itself,

$$\Delta M_{\text{crit},e} = -\int_0^{\tau_{\text{end}}} \rho(s)\, w(s)\, ds, \quad w(s) = \frac{Wz_{CoM} \displaystyle\int_s^{\tau_{\text{end}}} \sinh(C_2\tau)\cosh\big(C_2(\tau - s)\big)\, d\tau}{J_P C_2 \| \sinh(C_2\tau)\|^2}. \tag{D.13}$$

Step 4: three properties of the weight. All three are checked in code (`error_budget.py --check`).

(i) $w \geq 0$: the integrand is a product of two hyperbolic functions of non-negative arguments.

(ii) $\int_0^{\tau_{\text{end}}} w = 1$, *exactly.* Integrate the numerator of (D.13) over s and exchange the order once more — the same triangle again —

$$\int_0^{\tau_{\text{end}}}\int_s^{\tau_{\text{end}}} \sinh(C_2\tau)\cosh\big(C_2(\tau - s)\big)\, d\tau\, ds = \int_0^{\tau_{\text{end}}} \sinh(C_2\tau) \left[\int_0^{\tau} \cosh\big(C_2(\tau - s)\big) ds\right] d\tau.$$

The inner integral is elementary, $u = \tau - s$ turning it into $\int_0^\tau \cosh(C_2 u) du = \sinh(C_2\tau)/C_2$, so the whole numerator becomes $\|\sinh(C_2\tau)\|^2/C_2$ — the very norm sitting in the denominator of (D.13). Hence

$$\int_0^{\tau_{\text{end}}} w = \frac{Wz_{CoM}\, \|\sinh\|^2/C_2}{J_P C_2 \|\sinh\|^2} = \frac{Wz_{CoM}}{J_P C_2^2} = 1$$

by $J_P C_2^2 = Wz_{CoM}$, the identity that removes the inertia everywhere else in this document. Neither the window length nor any constant survives. `error_budget.py --check` evaluates (D.13) at $\rho \equiv 1$ and returns -1.0005 to -1.0001 across five operating points, which is quadrature error on an identity rather than evidence for it.

(iii) w is non-increasing in s: as s grows both the integration range and the cosh factor shrink.

Step 5: what (i) and (ii) alone give. A non-negative weight of unit mass can do no more than pick out the largest value of what it averages, so

$$\big|\Delta M_{\text{crit},e}\big| \;\leq\; \rho_{\max} \int_0^{\tau_{\text{end}}} w \;=\; \rho_{\max}.$$

This is already a result — a uniform unmodelled moment displaces the identified threshold one-for-one, as a consequence rather than an assumption — but it is the weak form, and over the box it is 2.70 mm of offset, larger than the accuracy actually achieved.

Step 6: what (iii) adds. The forcing rises across the window while the weight falls, so the two are oppositely monotone and Chebyshev's inequality (D.4) applies a second time, in the same direction as the first:

$$\Big|\int_0^{\tau_{\text{end}}} \rho\, w\Big| \;\leq\; \frac{1}{\tau_{\text{end}}} \int_0^{\tau_{\text{end}}} \rho \;\cdot \int_0^{\tau_{\text{end}}} w \;=\; \bar{\rho}\cdot 1 \;=\; \bar{\rho}.$$

The maximum has been replaced by the window average, and by (ii) no kernel gain is left over to multiply it.

Step 7: bound the average. $\bar{\rho}$ was computed at (95) as $R_\varphi \rho_{\varphi,\max} + R_{GE}\rho_{GE,\max}$, and the two reduction factors are bounded there by $\frac{1}{7}$ and $\frac{1}{5}$. Chaining Steps 6 and 7,

$$\big|\Delta M_{\text{crit},e}\big| \;\leq\; \bar{\rho} \;\leq\; \frac{\rho_{\varphi,\max}}{7} + \frac{\rho_{GE,\max}}{5}, \tag{97b}$$

the right half of (97). Note the direction of the second inequality: $\bar{\rho}$ is the average, and the right-hand side bounds it — it is not a definition of $\bar{\rho}$.

step	what changes	why it is allowed
1–2	$e_\omega \to$ Duhamel double integral	(D.2), the exact solution
3	double integral $\to \int \rho\, w$	Fubini on the triangle
4	—	$w \geq 0$, $\int w = 1$, $w \downarrow$
5	$\int \rho w \to \rho_{\max}$	(i) and (ii)
6	$\rho_{\max} \to \bar{\rho}$	(iii) plus Chebyshev (D.4)
7	$\bar{\rho} \to \rho_\varphi/7 + \rho_{GE}/5$	(95)

Why the constants differ from the left half. The left half asks how far the rate curve is displaced, so the kernel integral Ψ multiplies $\bar{\rho}$ and the products ΨR climb to $\frac{1}{2}$ and 1. The right half asks how far the *minimiser* moves, and the minimiser is found by correlating against $\sinh(C_2\tau)$ — a normalised operation, $\int w = 1$, with no kernel gain left over. So there $\bar{\rho}$ enters bare, and the constants are the R's themselves, $\frac{1}{7}$ and $\frac{1}{5}$. The threshold bound is the stronger of the two by roughly a factor of four, which is why it, and not the relative rate bound, is the number carried into the offset budget.

7. The threshold bound by hand

The manuscript reports only the threshold bound, and that branch uses neither Ψ nor x: the weight of (D.13) is normalised, so $\bar{\rho}$ enters bare and the constants are the R's themselves. What is left is arithmetic. No hyperbolic function appears below, C_2 never enters, and the window length is never solved for — $\frac{1}{7}$ and $\frac{1}{5}$ are suprema over *all* x, so no x has to be known to use them.

Given the weight W, the height z_{CoM}, the arm $a = l_p + p_{\text{off}}$, the mass m, a CAD inertia J_{CoM}, the ramp rate $\dot{M}$, the tilt cap $\varphi_{\max}$ and the ground-effect slope β_M, six lines:

(1) $J_P \leq J_{CoM} + m\big(z_{CoM}^2 + a^2\big)$ — (82) at its upper end

(2) $\Delta M_{\text{win}} \leq \big(6\varphi_{\max} J_P \dot{M}^2\big)^{1/3}$ — (93)

(3) $\rho_{\varphi,\max} = \frac{1}{2} W a\, \varphi_{\max}^2$ — the gravity remainder at the cap

(4) $\rho_{GE,\max} = |\beta_M|\, \Delta M_{\text{win}}\, \varphi_{\max}$

(5) $|\Delta M_{\text{crit}}| \leq \rho_{\varphi,\max}/7 + \rho_{GE,\max}/5$ — (97b)

(6) divide by $W_{\min}$ for the equivalent offset.

Over the geometric box of Sec. VI-D, with $W = 31.59$ N, $z_{CoM} = 0.30$ m, $m = 3.221$ kg, $J_{CoM} = 0.0537\ \text{kg}\,\text{m}^2$, $\varphi_{\max} = 10°$, $\beta_M = 0.0345\ \text{rad}^{-1}$ and $W_{\min} = 30.08$ N:

axis	$\dot{M}$ N m/s	J_P kg m^2	ΔM_{win} N m	$\rho_{\varphi,\max}$ mN m	$\rho_{GE,\max}$ mN m	$\lvert\Delta M_{\text{crit}}\rvert$ mN m	offset mm
roll ($a = 0.160$)	0.10	0.4260	0.165	76.98	0.99	11.20	0.372
	0.45	0.4260	0.449	76.98	2.70	11.54	0.384
	1.20	0.4260	0.863	76.98	5.20	12.04	0.400
pitch ($a = 0.130$)	0.10	0.3980	0.161	62.55	0.97	9.13	0.304
	0.45	0.3980	0.439	62.55	2.64	9.46	0.315
	1.20	0.3980	0.844	62.55	5.08	9.95	0.331

Three readings.

The inertia hardly matters. J_P enters only through ΔM_{win}, which enters only through $\rho_{GE,\max}$, which is 2–10% of the total; and it enters as a cube root. Halving J_P moves the bound by -1.8%, doubling it by +2.2%. A crude inertia is therefore good enough, which is worth knowing because (82) is the one input here that is not directly measured.

Line (5) is where the work went. Properties 1 and 2 of the weight alone — $w \geq 0$ and $\int w = 1$ — already give $|\Delta M_{\text{crit}}| \leq \rho_{\max} = \rho_{\varphi,\max} + \rho_{GE,\max}$, without Chebyshev and without any R. That is 2.70 mm at worst over the box, which *exceeds* the 1.64 mm validation RMS: a guarantee weaker than the measured performance, and so of no use. The factor of seven that $\frac{1}{7}$ and $\frac{1}{5}$ supply is what brings it to 0.40 mm and clears the measurement by 4.2×.

The rate bound is not needed for this. The same construction also bounds the relative rate deviation, with $\frac{1}{2}$ and 1 in place of $\frac{1}{7}$ and $\frac{1}{5}$ and a factor $\Psi = x\coth(x/2)$; that is Secs. 4–5, it is weaker by about a factor of four, and no number above depends on it.

8. A worked example, checkable by hand

On which x to use. Two values circulate below and they must not be confused. The cubic shortcut (92) supplies $\bar{x} = (6\Lambda_{\max})^{1/3}$, an *upper bound* on x obtained by replacing $\Lambda(x)$ with its leading term $x^3/6$. The exact value solves

$$\Lambda(x) \;=\; \Lambda_{\max} \;\triangleq\; \frac{\varphi_{\max} W z_{CoM} C_2}{\dot{M}}, \tag{D.14}$$

which has a unique root because Λ increases strictly from 0. The shortcut exists only to spare (97b) a transcendental solve it does not need — there x appears nowhere, and ΔM_{win} enters through a channel worth under a tenth of the total. Wherever x itself appears, (D.14) is solved instead, for two reasons. It is loose: at the slowest ramp Λ exceeds $x^3/6$ by 3.6×, and anything downstream of sinh exponentiates that. And it is not automatically valid: substituting an upper bound into $R(x)\sinh x$ presumes that product monotone, which is true but is not among the facts Sec. 5 proves, whereas solving (D.14) presumes nothing.

Both reduction factors close in finitely many terms, so the rows below need no quadrature:

$$\begin{aligned} R_\varphi(x) &= \frac{A_\varphi(x)}{x\Lambda(x)^2}, \qquad A_\varphi(x) = \tfrac{1}{4}\sinh 2x - \tfrac{1}{2}x - 2x\cosh x + 2\sinh x + \tfrac{1}{3}x^3, \\ R_{GE}(x) &= \frac{A_{GE}(x)}{x^2\Lambda(x)}, \quad A_{GE}(x) = x\cosh x - \sinh x - \tfrac{1}{3}x^3, \end{aligned} \tag{D.15}$$

each being $\int_0^x \Lambda^2$ and $\int_0^x u\Lambda$ integrated by parts against $\int \sinh^2 = \frac{1}{4}\sinh 2x - \frac{1}{2}x$ and $\int u \sinh u = u\cosh u - \sinh u$.

Take the roll axis at the fastest ramp, evaluated over the geometric box of Sec. VI-D rather than at any fitted constant: $W = 31.59$ N (fully ballasted, $m = 3.221$ kg), $z_{CoM} = 0.30$ m, $a = l_p + p_{\text{off}} = 0.160$ m, $\dot{M} = 1.20$ N m/s, $\varphi_{\max} = 10^\circ = 0.17453$ rad, $\beta_M = 0.0345$ rad^{-1}, $J_{CoM}^{\text{CAD}} = 0.0537$ kg m^2.

step	expression	value
geometry	$Wz_{CoM} = Wz_{CoM}$	$9.477\ \mathrm{N\,m}$
(82) lower	$J_P \geq m(z_{CoM}^2 + a^2)$	$0.3724\ \mathrm{kg\,m^2}$
(82) upper	$J_P \leq J_{CoM}^{\mathrm{CAD}} + m(z_{CoM}^2 + a^2)$	$0.4261\ \mathrm{kg\,m^2}$
	$C_2 \leq \sqrt{g z_{CoM}/(z_{CoM}^2 + a^2)}$	$5.045\ \mathrm{s}^{-1}$
tilt cap $\to \Lambda$	$\Lambda_{\max} = \varphi_{\max} W z_{CoM} C_2 / \dot{M}$	6.954
(D.14)	solve $\sinh x - x = 6.954$	$x = 2.9930$
	check: $\sinh(2.9930) - 2.9930$	6.9548
(92)	$\bar{x} = (6\Lambda_{\max})^{1/3}$, the shortcut	3.4685
(93)	$\tau_{\mathrm{end}} \leq (6\varphi_{\max} J_P/\dot{M})^{1/3}$	0.719 s
(93)	$\Delta M_{\mathrm{win}} = \dot{M}\tau_{\mathrm{end}}$	$0.863\ \mathrm{N\,m}$
channel	$\rho_{\varphi,\max} = \frac{1}{2} W a\, \varphi_{\max}^2$	$76.98\ \mathrm{mN\,m}$
(93)	$\rho_{GE,\max} = \lvert\beta_M\rvert \Delta M_{\mathrm{win}} \varphi_{\max}$	$5.20\ \mathrm{mN\,m}$
(D.15)	$\Lambda(x) = 6.9548$, $\Lambda(x)^2$	48.3690
(D.15)	$A_\varphi(x)$	17.2171
(D.6)	$R_\varphi(x) = 17.2171/[(2.9930)(48.3690)]$	0.1189
(D.15)	$A_{GE}(x)$	11.0389
(D.7)	$R_{GE}(x) = 11.0389/[(2.9930)^2(6.9548)]$	0.1772
(95)	$\bar{\rho} = R_\varphi \rho_{\varphi,\max} + R_{GE}\rho_{GE,\max}$ (against $\rho_{\max} = 82.2\ \mathrm{mN\,m}$)	$10.08\ \mathrm{mN\,m}$
(96)	$\Psi(x) = x\coth(x/2)$	3.309
(96)	$\Psi\bar{\rho}/\Delta M_{\mathrm{win}}$	3.86%
(97)	$(\frac{1}{2}\rho_{\varphi,\max} + \rho_{GE,\max})/\Delta M_{\mathrm{win}}$	5.06%
(97b)	$\rho_{\varphi,\max}/7 + \rho_{GE,\max}/5$	$12.04\ \mathrm{mN\,m}$
	$\div W_{\min}$	0.400 mm
(D.3)	envelope only, $\Psi\rho_{\max}/\Delta M_{\mathrm{win}}$	31.5%

Read the rows in two groups. Everything from (D.15) down to (96) is evaluated at the exact x of (D.14), so each is the value the run actually realises; the $\bar{x}$ row is carried only to show what the shortcut costs, and using it would have given $\bar{\rho} = 9.54$ instead of $10.08\ \mathrm{mN\,m}$ — an *under*statement, since R decreases. The (97) and (97b) rows are of a different kind again: they contain no x at all, so they hold whatever the window turns out to be. That is why (97b), and not the 3.86%, is what the offset budget inherits.

Two readings. First, (97) costs about a third over (96) — the price of dropping x from the statement — whereas the envelope form costs a factor of eight. Second, this is the *worst admissible* case: the box is evaluated at the 10° tilt cap, at the top of the z_{CoM} range, and with the inertia at whichever end of (82) is unfavourable for the quantity being bounded. Over the runs actually flown the excursions are 5–9° and the exact per-run propagation, which convolves the measured ρ instead of any envelope, gives $0.04\ \mathrm{mN\,m}$ in the median and 0.15 at worst — 0.001 to 0.004 mm of offset.

9. Summary of the chain

	step	what it buys
(D.2)	$e_\omega = \int \dot{h}_\varphi(\tau_{\text{end}} - s)\rho(s)ds$	the exact deviation
(D.3)	$\lvert\rho\rvert \le \rho_{\max}$, integrate the kernel	(90); too weak by $\sim 8\times$
(D.4)	Chebyshev, $\rho \uparrow$ against $\dot{h}_\varphi(\tau_{\text{end}} - \cdot) \downarrow$	$\rho_{\max} \to \bar{\rho}$
(D.6–7)	known shapes Λ^2 and $u\Lambda$	the ratios $\bar{\rho}/\rho_{\max}$
(D.6a)	$R = \int_0^1 f(xt)/f(x)dt$, $n_{\text{eff}} \uparrow$	(95): $R_\varphi \le \frac{1}{7}$, $R_{GE} \le \frac{1}{5}$
(96)	divide by $\omega_{\text{nom,end}}$, use $\Delta M_{\text{win}} = \dot{M}x/C_2$	$\Psi = x \coth(x/2)$; Wz_{CoM}, J_P gone
	$\Psi R_\varphi \le \frac{1}{2}$, $\Psi R_{GE} \le 1$	(97) left half; x gone
(D.12–13)	linearise the onset, Fubini, $w \ge 0$, $\int w = 1$	$\Delta M_{\text{crit}} = -\langle\rho\rangle_w$
	Chebyshev again, $\rho \uparrow$ against $w \downarrow$	(97) right half: $\frac{1}{7}$, $\frac{1}{5}$

Chebyshev is used twice, for the same reason both times: the forcing is concentrated at the end of the window and every kernel in the problem weights the end of the window least. The two uses are independent — the first converts a kernel integral, the second a normalised correlation — which is why the constants they leave behind differ.